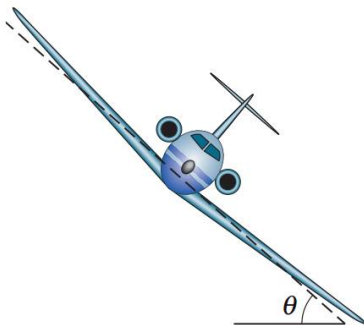


- 8 An airplane is flying in a horizontal circle at a speed of  $500 \text{ km h}^{-1}$  as shown below.



Assume that the aerodynamic lift force is perpendicular to the wing surface.

If its wings are tilted at an angle  $\theta = 40^\circ$  to the horizontal, what is the radius of the circle in which the plane is flying?

- A 1.3 km
- B 1.6 km
- C 2.0 km
- D 2.3 km

- 9 A satellite of mass  $50 \text{ kg}$  moves from a point where the gravitational potential due to the